

SLEEVEDRIFT · REQUIREMENTS

SleeveDrift requirements

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DOCUMENT CONTROL

Document ID	SVD-REQ-001	Version	0.4
Status	Draft	Date	2026-10-03
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/sleevedrift		
Source	docs/03-requirements.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.4	2026-10-03	Amish Chadha	Status after Amish's round 2 decisions (SVD-DDR-003, 1B, 2A, 3C) from SVD-CAL-001 v0.2; targets unchanged
0.3	2026-10-03	Amish Chadha	TRL 3 status from SVD-CAL-001 on the constructable design (SVD-DDR-002); targets unchanged
0.2	2026-10-03	Amish Chadha	TRL 2 update; concept status for every requirement; R11 (overload limit) and R12 (gas monitoring) added under Amish's pre-approval (SVD-DDR-001)
0.1	2026-09-30	Amish Chadha	Initial scaffold

SleeveDrift requirements

Ten of the twelve requirements are met on paper or by design; R3 (spoil haulage) is not met, R9 (shift durability) is at risk on paper, and the 600 mm stretch of R1 is open. Amish decided the three items on 2026-10-03 (SVD-DDR-003): a portal drill drive on each winch socket, with hand cranking as the fallback, doubles the haul to about 0.42 m³/h (R3, still short of 1 m³/h); the ropes are inspected every shift and a spare set is carried, with the TRL 4 endurance trial deciding whether a tougher rope is needed (R9); the 600 mm stretch stays open until the co-design partner answers (R1). Targets are unchanged from the scaffold; R11 and R12 were added at TRL 2 (SVD-DDR-001). The figures in the status column come from the calculation note SVD-CAL-001 v0.2 and are estimates on paper, not test results; every requirement is verified by test at TRL 4 or later.

Safety: These requirements describe equipment used by people inside a rescue pipe under collapse debris. Meeting them on paper does not make the kit safe to use; it is published as an open engineering reference, not certified rescue, mining or confined-space equipment.

Table 1. Requirements, with their status at TRL 3.

ID	Requirement	Target	Verification (TRL 4 or later)	Status at TRL 3
R1	Fits the pipe	Every part passes through and assembles inside an 800 mm nominal pipe; hood and ring variants for 600 mm (stretch)	Fit trial in pipe sections	Met on paper for 800 mm: every piece passes a 776 mm circle in a 780 mm bore. 600 mm stretch open: kept as a stretch until the co-design partner says whether 600 mm pipes are used for rescue (SVD-DDR-003)
R2	Hood protection	Hood carries a 5 kN (1,124 lbf) distributed load with no contact with a test head form	Load test on CalRig	Met on paper: 5.5 MPa in the crown, tip deflection under 0.1 mm
R3	Spoil haulage	At least 1 m ³ (35 ft ³) of loose spoil per hour moved 60 m (197 ft) by a two-person portal crew	Timed trial in a 60 m pipe run	Not met: about 0.42 m ³ /h with the portal drill drive (SVD-DDR-003); 0.22 m ³ /h by hand, the fallback
R4	Haul force	Portal crank force under 200 N (45 lbf) with loaded sledges over 60 m	Force gauge during trial	Met on paper: 92 N at the start in high gear, 30 N in low gear, when cranked by hand; the drill drive takes the haul in normal use
R5	Fresh air at the face	At least 1 m ³ /min (35 ft ³ /min) of fresh air delivered at the face over 60 m of duct (target, to be checked against confined-space guidance)	Anemometer at duct outlet	Met on paper: about 15 m ³ /min at the face
R6	Casualty evacuation	A 100 kg (220 lb) manikin moved 60 m through the pipe in under 5 minutes	Timed drill	Met on paper: about 3.5 min with three people hauling hand over hand
R7	Rapid set-up	Kit installed and hauling within 2 hours of arriving at the portal	Timed deployment exercise	Met on paper: about 1.6 h estimated
R8	No power in the pipe	Zero electrically powered items inside the pipe except battery lamps and signal devices	Design review	Met by design: battery lamps, gas monitors (alarm devices) and sound-powered telephones only; the battery drill drives stay at the portal
R9	Shift durability	72 hours of continuous operation without failure of rope, pulleys or sledges	Endurance trial	At risk on paper: rope abrasion over about 250 round trips at the drill-drive rate cannot be shown on paper. Met in practice if a rope swap counts as maintenance: ropes inspected at every shift change against a wear gauge, a spare set carried (SVD-DDR-003); the TRL 4 endurance trial decides whether an abrasion-resistant rope is needed
R10	Transportable	Whole kit in cases no heavier than 40 kg (88 lb) each, moveable by road and by hand to the portal	Weighing and handling trial	Met: heaviest package 34.8 kg; kit about 326 kg in 14 packages with the spare ropes and drill drives

ID	Requirement	Target	Verification (TRL 4 or later)	Status at TRL 3
R11	Overload limit	Rope tension limited to 2.5 kN (562 lbf) by a release at each end of the train or stretcher; every part sized on that limit	Release test of the swivel pins; proof load on CalRig	Met by design: rope factor 6.5 on the release load
R12	Air monitoring	Oxygen, carbon monoxide, hydrogen sulphide and flammable gas monitored continuously at the face and at the portal, with alarms	Design review; bump test before each shift	Met by design: two four-gas monitors in the kit

■ Assumptions

- The rescue pipe is steel of 800 mm outside diameter with a 10 mm wall (780 mm bore), as at Silkyara; the fit trial confirms the bore and the weld beads.
- Experienced hand miners or trained rescue crews will be available to use the kit.
- A blower and some power are available at the portal; nothing inside the pipe uses mains power.
- The pipe is pushed forward by other means while crews dig; SleeveDrift does not push pipe.
- The pipe is horizontal or on a slight grade, so the train does not run away when a winch is eased.